Gary: database and mobile handoff

From Wisani, 17 October 2026 (updated 18 Oct for the Unit 4 notes) · Spanner (Riese Auto-Clinic) · Phase 3 starts now, due Monday 2 November

Short version. I checked GitHub: nothing has changed in `mobile/` or the database files on `main` since 28 September, so none of the earlier tweaks have landed yet. Everything below is still open. In order of urgency:

1. **Revoke the leaked Firebase key** (by Mon 19 Oct).
2. **Run five SQL files** in the Supabase SQL editor, then send me the results of three check queries (by Wed 21 Oct).
3. **Mobile changes**: Send on WhatsApp button (Mr G asked for it at the presentation), make the app build for the web so we can host it on Vercel, the tracking patch, the client colours (by Sat 24 Oct).

Tell me in the group when each one is done, or if anything doesn't work. If you're stuck on the SQL, give me temporary access and I'll run it with you.

1 Revoke the leaked Firebase key

The Firebase service-account file was removed from the repo on 28 Sept, but it's still in the git history, and the repo is public. Deleting the file doesn't stop anyone using the key; only revoking it does.

1. Google Cloud console → the Firebase project → **IAM & Admin → Service Accounts**.
2. Open the `firebase-adminsdk` account → **Keys** → delete the key whose ID matches the leaked file.
3. Create a new key **only if** the Edge Function `send-phase-notification` needs one. Put it in Supabase as an Edge Function secret, never in the repo or in `.env`.
4. Test: change a booking's phase and check the phone still gets the push.

2 Database: five SQL files, in this order

All are in `supabase/migrations/` on GitHub. Open each file, paste the whole thing into **Supabase → SQL Editor**, run it. Every one is safe to run twice.

File	What it fixes
0009 drop old add_booking_part	"Add part" on the web fails with error 42804. The first version of the function (text user id) is still in the database next to the fixed one, and Postgres calls the old one. 0009 drops the old one.
0004 atomic stock decrement	Re-creates the correct `add_booking_part` (uuid user id): stock decrement, booking part and stock movement in one step, so stock can't be lost if a call fails halfway.
0008 lock staff functions	**Security.** On 7 Oct I found that `update_booking_phase` runs *without a login*, using the anon key that ships inside the APK. Anyone with a booking id could change its phase (which pushes a notification to the customer) and get back the customer's name, phone and email. 0008 makes the staff functions signed-in only. The customer lookup stays public.
0010 bookings.created_by	Ms Mbele's ERD question at the presentation: "who is logged in doing this?" Adds `created_by` to bookings, filled in automatically with the signed-in staff member. No app changes needed.
0011 finalize/reopen admin-only *(new 18 Oct)*	The Unit 4 notes say access must be enforced by the database, "rather than relying only on hiding buttons". Delete already is (0007); finalize and reopen were only hidden buttons. 0011 adds the same admin check as 0007 on any status change. Tested on a local Postgres: mechanic refused, admin allowed, phase changes unaffected. Every web admin needs `role = 'admin'` in `profiles`.

Then: check 0006 against the real database

`0006` is **not** to be run. It's my reconstruction of the tables and functions you built in the dashboard (login, phases, reference code, push tokens), so the repo and the ERD show the real schema. Please run `supabase db pull` (or copy the real definitions from the dashboard) and fix anything in 0006 marked `[CHECK]` that differs.

Check queries: run these four and send me a screenshot

```
-- 1. Should say FALSE once 0008 is applied (anon can no longer change phases)
select has_function_privilege('anon', 'public.update_booking_phase(uuid,text)', 'execute');

-- 2. Should return ONE row, with a uuid p_user_id (the old text version is gone)
select pg_get_function_identity_arguments(oid)
from pg_proc where proname = 'add_booking_part';

-- 3. Should return one row: created_by
select column_name from information_schema.columns
where table_schema = 'public' and table_name = 'bookings' and column_name = 'created_by';

-- 4. Should return one row: the 0011 trigger
select tgname from pg_trigger where tgname = 'bookings_admin_status_change';
```

3 Mobile: Send on WhatsApp button

Mr G at the presentation: a button that sends the customer their reference code on WhatsApp, to be shown at the final presentation. The web app has it already (booking page, `web/Services/CustomerWhatsApp.cs`). Please add the same to `booking_detail_screen.dart`, next to the *Reference:* line.

It uses WhatsApp's free click-to-chat link (`https://wa.me/<number>?text=...`), not the paid WhatsApp Business API, so no keys, no Meta account, no cost. It opens WhatsApp with the message typed in; staff check it and tap Send.

1. Add the package: `flutter pub add url_launcher`

2. New file `lib/features/bookings/data/customer_whatsapp.dart`. Same rules and wording as the web version, so the customer gets the same message from either app:

```
import '../models/booking_phase.dart';
import '../models/booking_status.dart';
import '../models/vehicle_booking.dart';

/// Same wording as the web app's /Track page and the push notification.
const _phaseDetail = {
  BookingPhase.bookedIn: 'Your vehicle is with us and being checked.',
  BookingPhase.laborInProgress: 'Our mechanics are working on your vehicle.',
  BookingPhase.reassembly: 'Putting everything back together and testing.',
  BookingPhase.readyForPickup: 'All done. You can collect your vehicle.',
  BookingPhase.finalized: 'Booked out and invoiced. Thank you!',
};

/// "082 123 4567" -> "27821234567". Null if we can't be sure, so WhatsApp
/// asks staff to pick the chat instead of messaging a wrong number.
String? toWhatsAppNumber(String? phone) {
  if (phone == null || phone.trim().isEmpty) return null;
  final trimmed = phone.trim();
  var digits = trimmed.replaceAll(RegExp(r'\D'), '');
  if (digits.startsWith('00')) {
    digits = digits.substring(2);
  } else if (!trimmed.startsWith('+') && digits.length == 10 && digits[0] == '0') {
    digits = '27${digits.substring(1)}';
  }
  return digits.length >= 10 && digits.length <= 15 && digits[0] != '0' ? digits : null;
}

const publicSiteUrl = 'https://riese-auto-clinic.vercel.app';

String whatsAppMessage(VehicleBooking b) {
  final first = b.customerName.trim().split(RegExp(r'\s+')).first;
  final hi = first.isEmpty ? 'Hi' : 'Hi $first';
  final car = '${b.brand} ${b.model}'.trim() +
      (b.registration.isEmpty ? '' : ' (${b.registration})');
  final code = b.referenceCode;

  if (b.status != BookingStatus.checkedIn || b.phase == BookingPhase.finalized) {
    return '$hi, your $car has been booked out (invoice ${b.invoiceNumber}). '
        'Thank you for choosing Riese Auto-Clinic!';
  }
  final text = b.phase == BookingPhase.bookedIn
      ? '$hi, your $car is booked in at Riese Auto-Clinic. Your reference code is $code.'
      : '$hi, an update on your $car from Riese Auto-Clinic: '
        '${b.phase.displayName}. ${_phaseDetail[b.phase]} Reference code: $code.';
  return '$text Track it any time: $publicSiteUrl/Track?code=${Uri.encodeComponent(code)}';
}

Uri whatsAppLink(VehicleBooking b) {
  final text = Uri.encodeComponent(whatsAppMessage(b));
  final number = toWhatsAppNumber(b.customerPhone);
  return Uri.parse(number == null
      ? 'https://wa.me/?text=$text'
      : 'https://wa.me/$number?text=$text');
}
```

Not compiled: I don't have Flutter here. The field names match `VehicleBooking` on `main`; run `flutter analyze` after adding it.

3. The button in `booking_detail_screen.dart`, under `Text('Reference: ${booking.referenceCode}')`:

```
import 'package:url_launcher/url_launcher.dart';
import '../data/customer_whatsapp.dart';
...
FilledButton.icon(
  style: FilledButton.styleFrom(backgroundColor: const Color(0xFF25D366)),
  icon: const Icon(Icons.chat),
  label: const Text('Send on WhatsApp'),
  onPressed: () => launchUrl(whatsAppLink(booking),
      mode: LaunchMode.externalApplication),
),
```

4. Android 11+ only opens links to other apps if they're declared. In `android/app/src/main/AndroidManifest.xml`, inside `<manifest>`:

```
<queries>
  <intent>
    <action android:name="android.intent.action.VIEW" />
    <data android:scheme="https" />
  </intent>
</queries>
```

Test: book in a car with your own number as the customer's, tap the button, send it, check it arrives with the code and the link. Change the phase and send again; finalize and send again (thank-you with invoice number).

4 Make the app build for the web (for Vercel)

We're putting the Flutter app online at **riese-auto-clinic-app.vercel.app**, next to the web app at **riese-auto-clinic.vercel.app**, so markers and customers can open it on any phone without the APK. Full plan: `project-docs/phase-3/hosting-on-vercel.md`.

1. Add the web platform (the repo only has `android/`): `cd mobile` then `flutter create --platforms=web .`
2. `main.dart` already skips Firebase on the web. But `PushNotificationService.registerForNotifications` doesn't, so it throws on the web when a customer enters their code. Add at the top of the method:

   ```
   import 'package:flutter/foundation.dart' show kIsWeb;
   ...
   Future<void> registerForNotifications(String referenceCode) async {
     if (kIsWeb) return; // push needs Firebase, which we only set up on Android
     ...
   ```

3. `flutter build web --release`, then `flutter run -d chrome` to test: login, bookings, book in, customer lookup with a code.
4. Push it; I'll do the Vercel deploy from `mobile/build/web` with you.

> **The `.env` file becomes public on the web.** `pubspec.yaml` bundles `.env` as an asset, and in a web build anyone can download it. That's fine for `SUPABASE_URL` and `SUPABASE_ANON_KEY` (they're public by design; the RLS rules protect the data, which is why 0008 matters). **Never put anything else in that file**: no service-role key, no Firebase server key.

5 The tracking patch, add part, and the client colours

- **Customer tracking patch** (sent 7 Oct, branch `proposal/mobile-customer-tracking`): the courier-style status screen, plus the fix that stops notification setup from blocking the status screen. It compiled with `flutter analyze` clean. Please review and apply it, or tell me what to change. Ask me to re-send it if you don't have it any more.
- **Add part in one step** *(new 18 Oct)*: mobile adds a part with three separate calls (part, stock, movement). If one fails halfway, stock is wrong. Once 0009 and 0004 are run, call the same function the web uses:

```
final rows = await _client.rpc('add_booking_part', params: {
  'p_booking_id': bookingId,
  'p_stock_item_id': stockItemId,
  'p_qty': quantity,
  'p_unit_price': item.sellPrice,          // price at the time, like the web
  'p_user_id': _client.auth.currentUser!.id,
  'p_reference': booking.registration,     // shows in stock_movements
}) as List;
if (rows.isEmpty || rows.first['ok'] != true) {
  throw Exception('Not enough stock'); // nothing was changed
}
```

  It returns `ok` and the new `quantity_on_hand`; if `ok` is false, nothing was saved.

- **Client colours** (Michael chose gold and green, or gold, green and orange; the web app already uses them). In `main.dart`'s `MaterialApp`:

```
theme: ThemeData(
  colorScheme: ColorScheme.fromSeed(
    seedColor: const Color(0xFF2E7D4F),   // green
    primary: const Color(0xFF2E7D4F),
    secondary: const Color(0xFFC9A227),   // gold
    tertiary: const Color(0xFFD9601A),    // orange (warnings, "in progress")
  ),
  scaffoldBackgroundColor: const Color(0xFFF6F2E8),
),
```

  And change the title from `'Stock System Riese Auto Clinic'` to `'Riese Auto-Clinic'`.

6 A decision for Tuesday's meeting: one labour ledger

Mobile records labour in `booking_labor`; the web app records it in `jobs` (hours × rate). Same bookings, two tables, so the two invoices can show different totals. **My proposal:** keep `booking_labor`, because mobile, the customer tracking lookup and the push already use it. I move the web app over; `jobs` keeps only the job type. Have a think, and tell me if your code reads labour anywhere I haven't listed.

7 Checklist: tick and send back

	Item	By
☐	Firebase key revoked; push still works	Mon 19 Oct
☐	0009, 0004, 0008, 0010, 0011 run; screenshot of the 4 check queries sent	Wed 21 Oct
☐	0006 checked against the real database (`supabase db pull`)	Wed 21 Oct
☐	Web "Add part" works after 0009 (I'll test it; tell me when it's run)	Wed 21 Oct
☐	Mobile Send on WhatsApp works on a real phone	Sat 24 Oct
☐	Web platform added, `kIsWeb` guard, `flutter build web` works	Sat 24 Oct
☐	Tracking patch applied (or feedback on it)	Sat 24 Oct
☐	Client colours and app title	Sat 24 Oct
☐	Mobile add part through `add_booking_part`	Sun 25 Oct
☐	Supabase Table Editor screenshots for the evidence (keys and personal data hidden)	Tue 27 Oct
☐	Everything pushed to GitHub (a branch and pull request is fine)	Sat 24 Oct
☐	Your Individual Contributions rows in the Phase 3 tracker	Thu 29 Oct

Why now: Phase 3 is due **Mon 2 Nov** (we submit on **Fri 30 Oct**), presentations are 2-6 Nov. Kea, Ofentse and Breyton test on the live addresses from 25 Oct, and Michael's acceptance test is planned for Sat 24 Oct. Both need the database fixes in first. Thanks, Gary.